

1. Primary Data Source & Query Interface

- **NCBI Protein** database & **UniProtKB** flatfile records
- Automated programmatic retrieval via **Biopython** (Entrez E-utilities)
- Dynamic request rate limiting with exponential backoff & retry mechanisms

2A. Oncogenic Dataset (Positive Class)

- **Inclusion Query:** *'oncogenic' OR 'tumorigenic' OR 'carcinogenic'*
 - Full GenBank flatfile retrieval (.gb / .gp)
- **Parsed Metadata:** Accession, Taxon ID, Length, FASTA
 - Literature citation linking (PMID, Journal, Year)

Label: Class 1 (Oncogenic)

2B. Non-Oncogenic Dataset (Negative Class)

- **Stage I:** Query for non-pathogenic fungal sequences
- **Stage II:** BLASTp homology screen ($\geq 90\%$ identity)
- **Stage III:** Exclusion query: *NOT ('oncogenic' OR 'tumorigenic' OR 'carcinogenic' OR 'cancer' OR 'oncoprotein' OR 'neoplasm')*

Label: Class 0 (Non-Oncogenic)

3. Quality Assurance, Sanitization & Redundancy Reduction

- Removal of sequence fragments shorter than **30 amino acids**
- Filtering of ambiguous / non-standard IUPAC characters (B, Z, X, J)
- **Sequence Deduplication (CD-HIT):** Clustered at **90% sequence identity threshold** ($-c\ 0.9\ -n\ 5$) to remove homologous redundancy and mitigate evaluation bias

4. Dataset Merging, Class Balancing & Stratified Partitioning

- **Balanced Benchmark Corpus:** $N = 4,808$ (2,404 Oncogenic / 2,404 Non-Oncogenic)
- Pseudorandom shuffling to eradicate sequential database extraction order bias
- **Stratified 80/20 Partitioning (Zero Data Leakage):**
 - **Training Subset (80%):** $n = 3,846$ (5-fold cross-validation & model training)
 - **Independent Test Subset (20%):** $n = 962$ (Unseen blind benchmark evaluation)